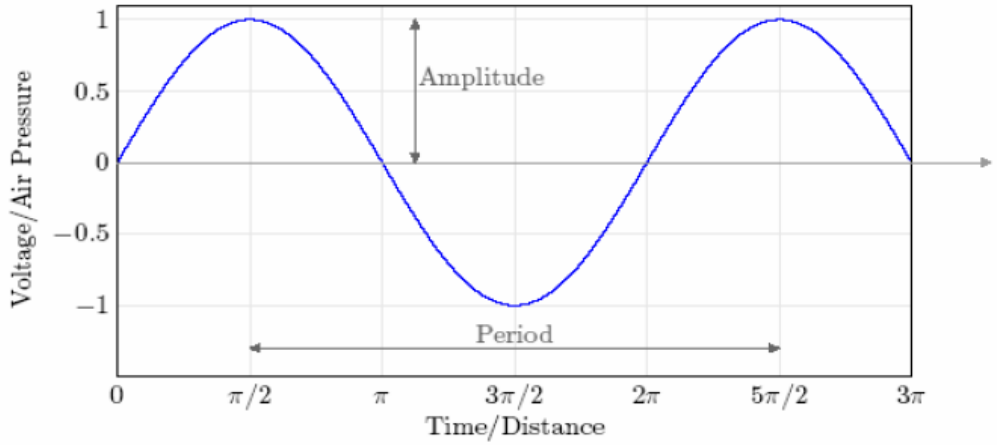
Signal Representations

EE3408

Signal Representation

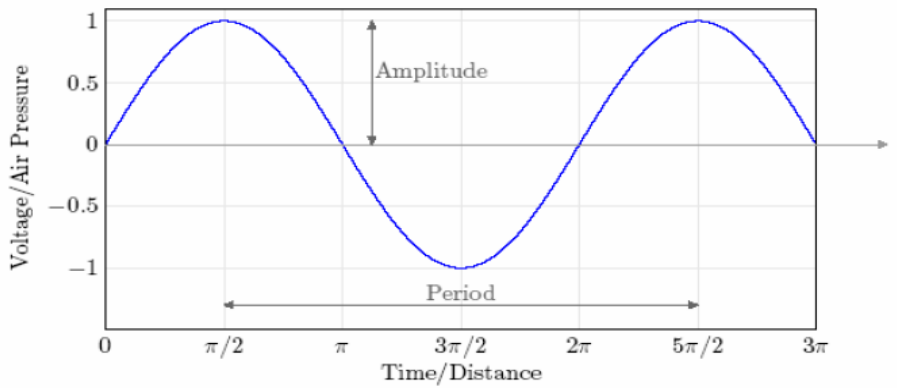
- A signal is some information somehow encoded as a wave.



Signal Representation ../2

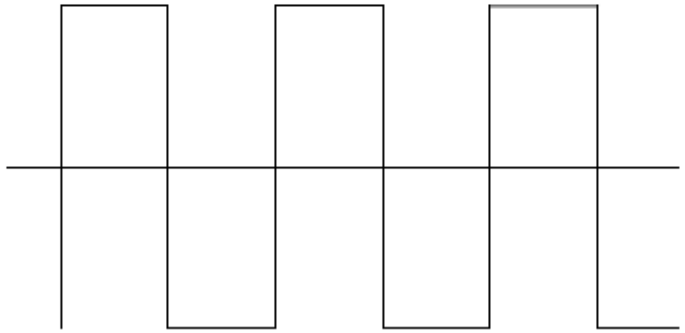
- Sine wave

$$v = V\sin(\omega t + \phi)$$



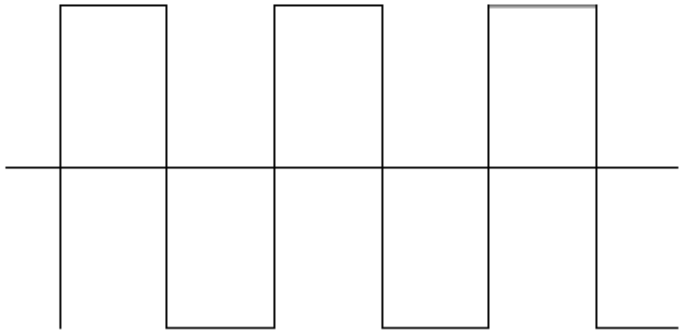
- Sinusoidal signals (sine or cosine waves)

- Non-sinusoidal signals (complex signals)
  - E.g. square wave
  - Can be represented by sum of a series of sine & cosine signals

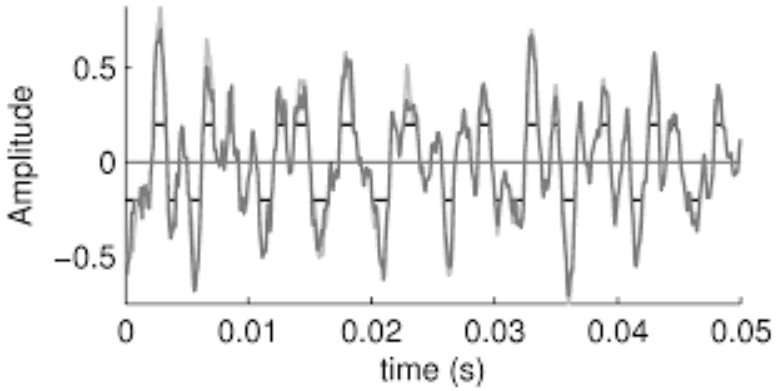


Signal Representation ../3

- Periodic waves
  - f(t) is periodic if there is T, such that f(t + T) = f(t)



- Non periodic signals (change continuously)
  - E.g. video & audio signals



Energy Signals

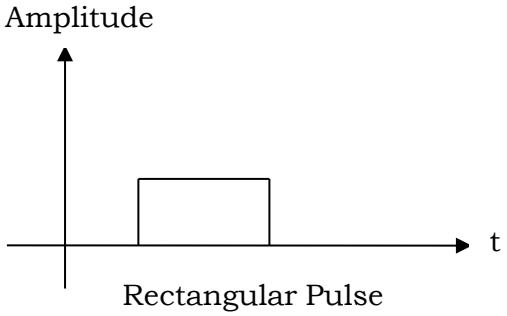
- pulse-like signal that usually exists for only a finite interval of time

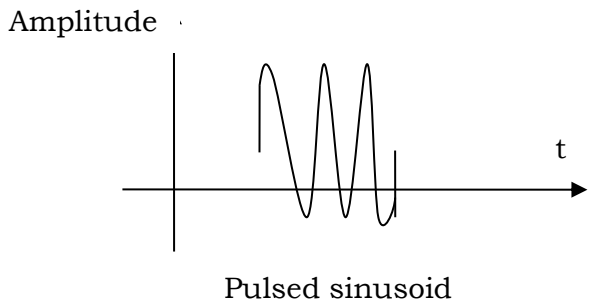
- Represented by

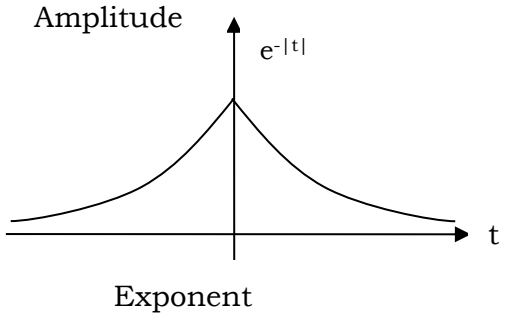
$$E = \int_{t_1}^{t_2} |f(t)|^2 \, dt$$

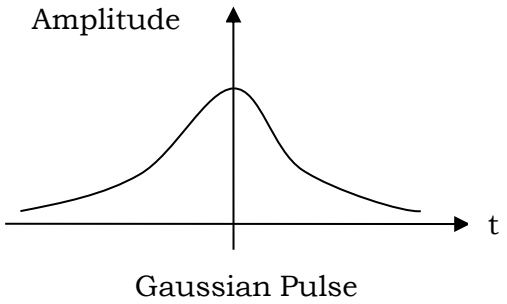
  - Is finite even when time interval becomes infinite, i.e. when

$$\int_{-\infty}^{\infty} |f(t)|^2 \, dt < \infty$$









Power Signal

- is the signal *f(t)* which has a finite average power even when the time interval becomes infinite

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} | f(t) |^2 \, dt$$

  - RHS remains finite, but non-zero when interval becomes infinite, i.e.

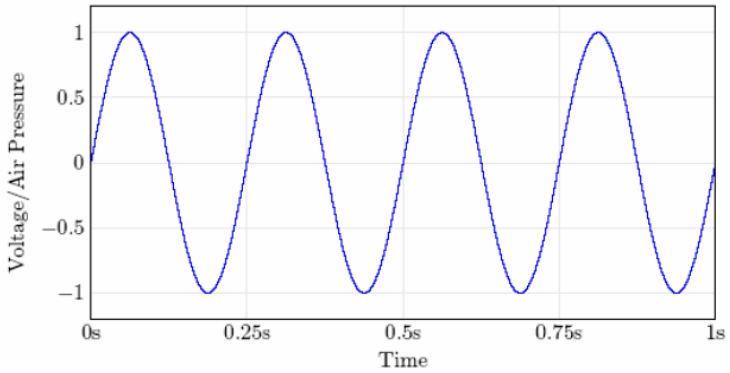
$$0 \leq \lim_{T \to \infty} \frac{1}{T} \int_{t_1}^{t_2} | f(t) |^2 \, dt < \infty$$

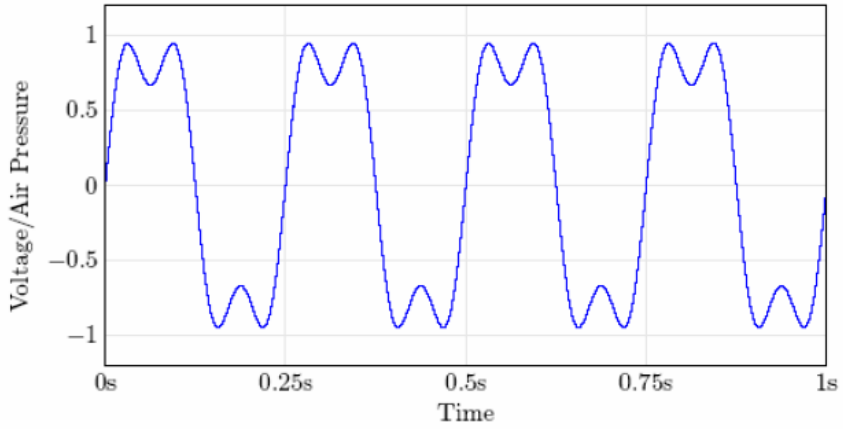
  - Is finite even when time interval becomes infinite, i.e. when

$$\int_{-\infty}^{\infty} | f(t) |^2 \, dt < \infty$$

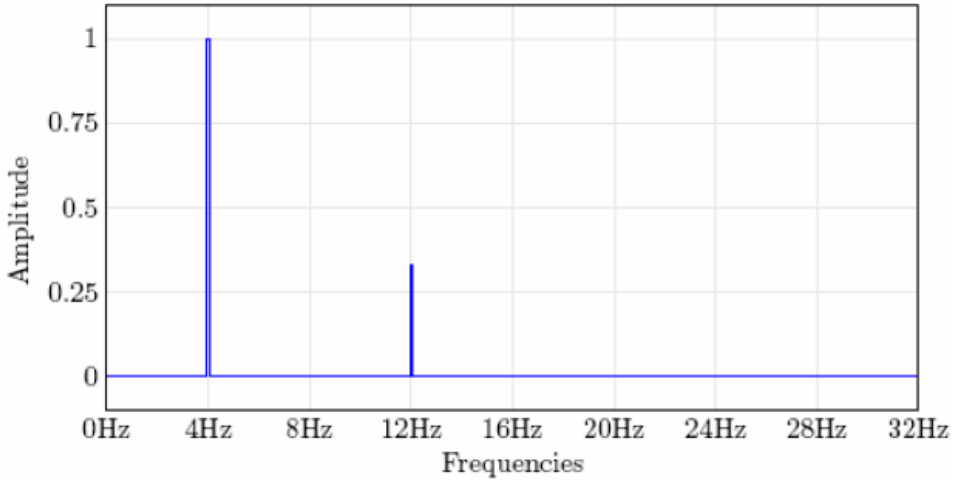
Time & Frequency Domains

- Time domain





- Frequency domain



Fourier Analysis

- is mathematical technique that is used to determine the amplitude and frequency of the sinusoidal components contained in any complex signal

- Fourier Series
  - Used to analyse periodic signals (square, sawtooth waves).

- Fourier Transform
  - Used to analyse non-periodic signals

Fourier Series

- Used to represent periodic waveforms by a series of sine and/or cosine waves
- When applied to a time-varying signal $f(t)$

$$f(t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) + B_n \sin(n\omega t)\right)$$

$$\omega_0 = 2\pi f_0; \qquad f_0 = \frac{1}{T} = \textit{fundamental frequency}$$

Fourier Series ../2

- Constants A_0, A_n and B_n are given by

$$A_0 = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\,dt$$

$$A_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\cos(n\omega t)\,dt$$

$$B_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(n\omega t)\,dt$$

  - A_0 term = the *DC component* (or dc offset) because it represents the average value of $f(t)$.

Fourier Series ../3

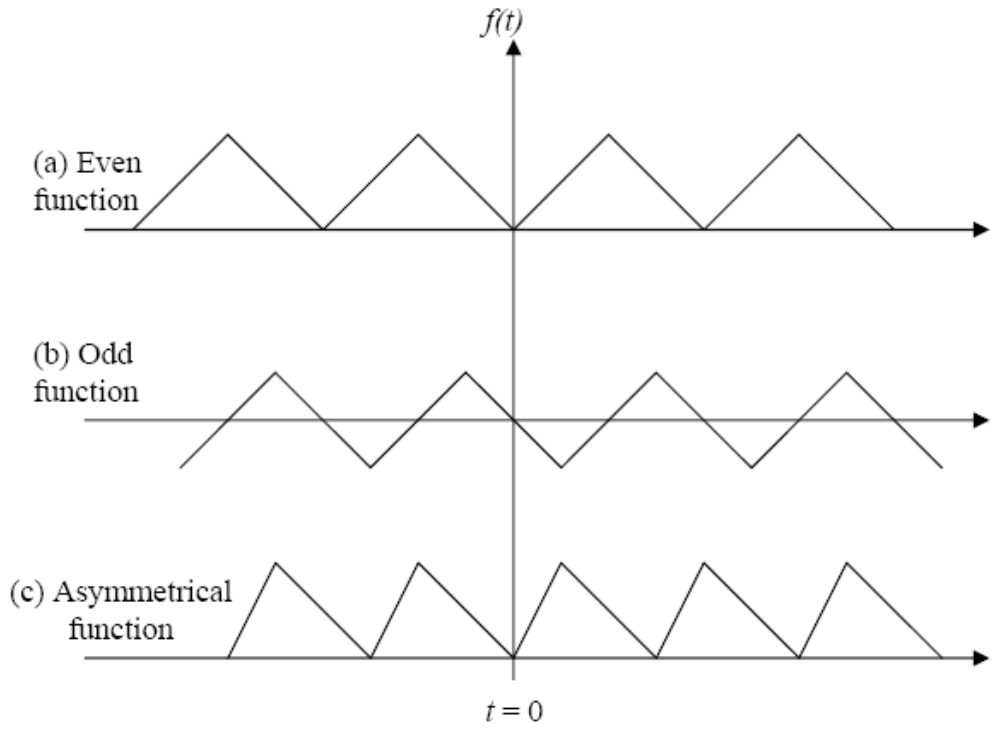
- A_1 and B_1
  - Are fundamental components because
  - their period is the same as the period of the complex signal

- Odd & Even functions
  - Waveform f(t) is even function if $f(t) = f(-t)$
    i.e. is symmetrical about $t = 0$

  - *if* $f(t) = -f(t)$
    - waveform is an odd function
    - Has skew symmetry about $t = 0$

Fourier Series ../4

- Function that has no symmetry about t = 0 axis
  - ➢is neither odd nor even



Fourier Series ../5

- If

$$f(t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) + B_n \sin(n\omega t)\right)$$

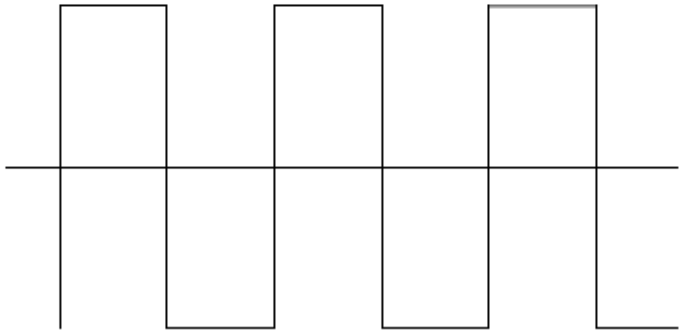
- then

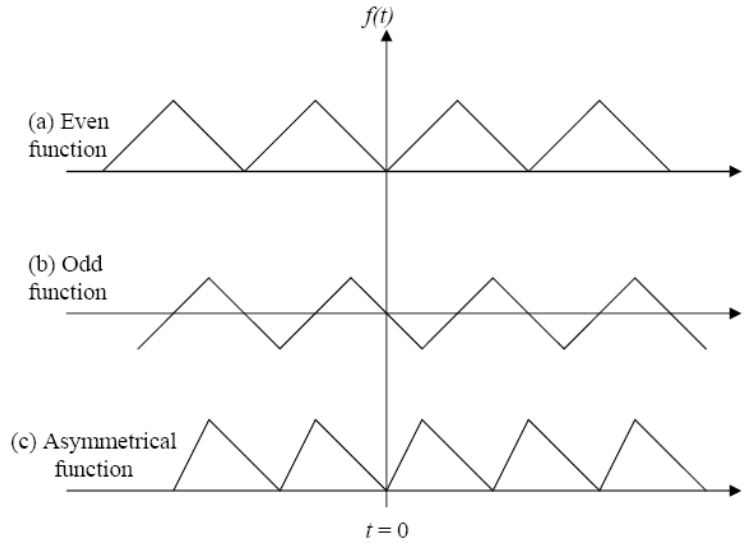
$$f(-t) = \frac{A_0}{T} + \frac{2}{T}\sum_{n=1}^{\infty}\left(A_n \cos(n\omega t) - B_n \sin(n\omega t)\right)$$

- If *f(t) = f(-t)*, this can only be true if $B_n = 0$.
  - i.e. Fourier series of an even function has cosine terms only.
- If *f(t) = -f(t)*, then $A_n = 0$.
  - i.e. Fourier series of an odd function has sine terms only.

Fourier Series ../6

- If *f(t)* has no symmetry about $t = 0$,
  - ➢Fourier series contains both sine and cosine terms.

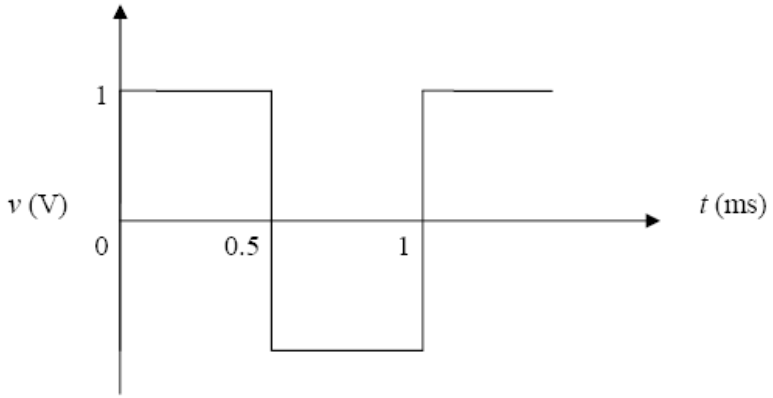




Fourier Series ../7

Example

- Fourier series of odd square wave



$$Amplitude = 1V; \qquad Period\ T = 1ms$$

$$Fundamental\ \ frequency\ \ f_0 = \frac{1}{T} = \frac{1}{10^{-3}} = 1kHz$$

$$B_n = \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t)\sin(n\omega t)\,dt = -\int_{-\frac{T}{2}}^{0} \sin(n\omega t)\,dt + \int_{0}^{\frac{T}{2}} \sin(n\omega t)\,dt = \frac{1}{n\pi f}\left[1 - \cos(n\pi)\right]$$

- *From Eqn:* $B_1 = \frac{2}{\pi f}$, $B_3 = \frac{2}{3\pi f}$, $B_5 = \frac{2}{5\pi f}$, and so on. However, $B_2 = B_4 = B_6 = \ldots = 0$

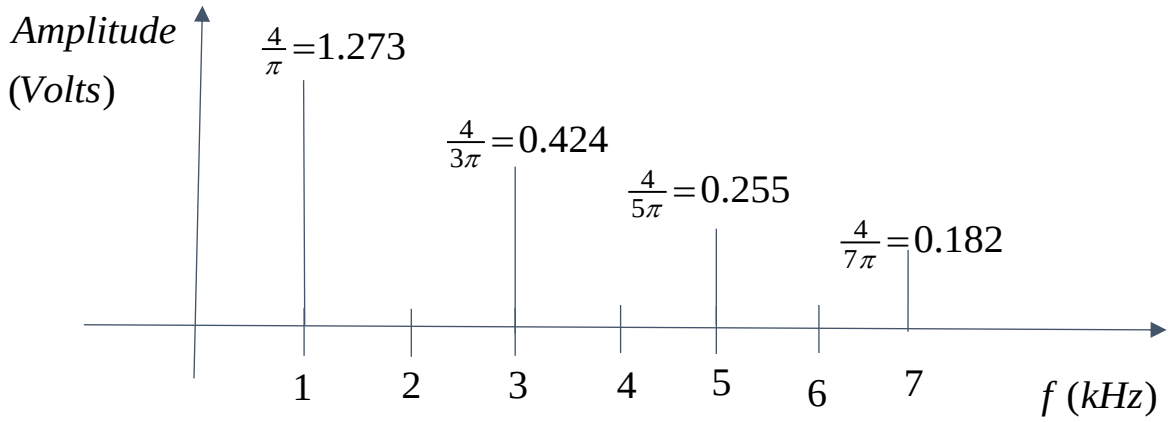
  - Substituting the values of B_1, B_3, etc, into F.S. equation gives

$$f(t) = \frac{4}{\pi}\left(\sin\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{5}\sin 5\omega t + \ldots\right)$$

Fourier Series ../

Example..

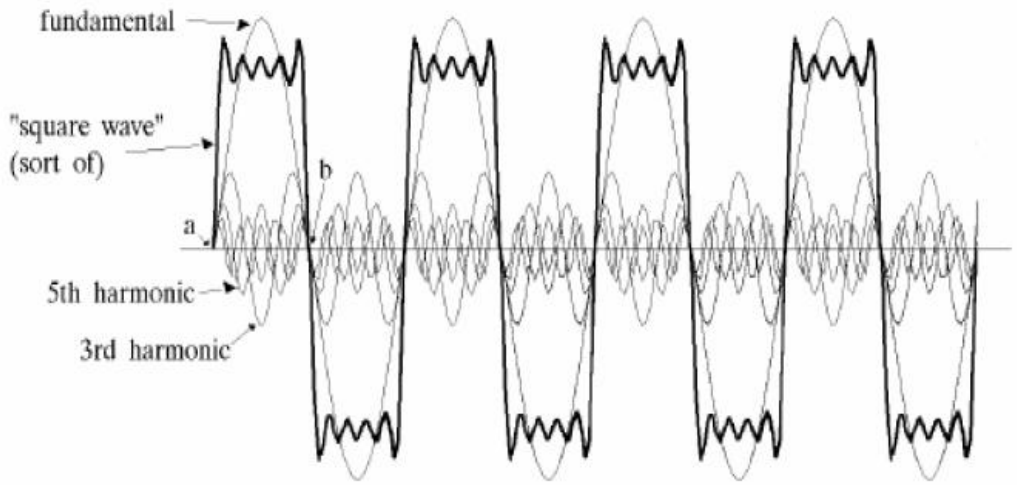
- where $\omega = 2\pi \, x \, 10^3$.
  - For 1 kHz component: amplitude of $\frac{4}{\pi}$ = 1.27 V,
  - For 3 kHz component: amplitude = ($\frac{4}{3\pi}$ = 0.424 V),
  - For 5 kHz: amplitude = ($\frac{4}{5\pi}$ = 0.255 V), and so on



Wave Synthesis

- Can be regarded as inverse of Fourier Analysis

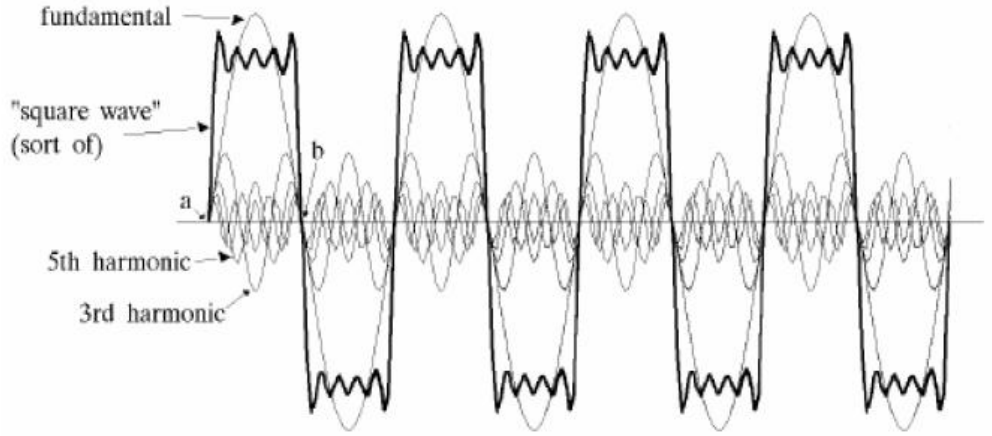
- Fourier
  - ➢ any periodic waveform can be synthesized by adding an infinite number of sinusoidal waves with specific amplitudes and frequencies



Wave Synthesis ../2

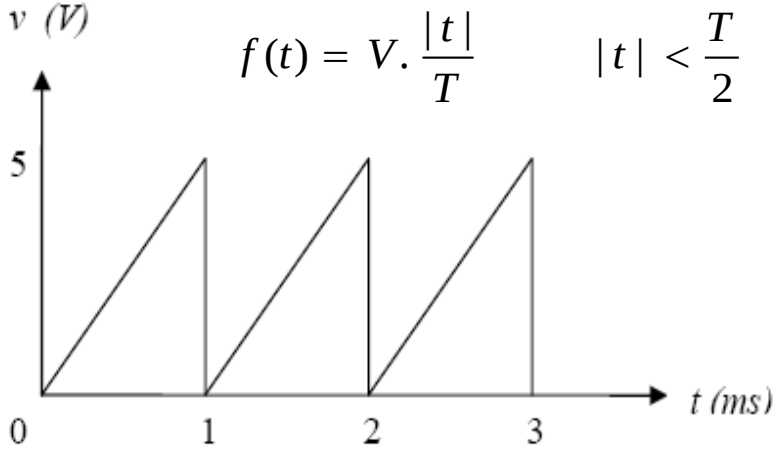
- This error in waveform synthesis is known as *Gibb's phenomenon*

- For square wave
  - The harmonics = odd multiples of the fundamental frequency
  - Amplitude of n harmonic $= \frac{1}{n}$ amplitude of f_0
  - Their phase has to be just right



Other Periodic Waves

- Positive going Sawtooth



Time domain

$f_0 = 1kHz; \quad Peak\ amplitude = 5V$

$$A_0 = \int_{\frac{-T}{2}}^{\frac{-T}{2}} f(t)dt = \int_0^T \frac{Vt}{T} dt = \frac{VT}{2}$$

$$A_n = \int_{\frac{-T}{2}}^{\frac{-T}{2}} f(t)\cos n\,\omega t dt = \int_0^T \frac{Vt}{T}\cos n\,\omega t dt$$

$$= \frac{V}{T(n\omega)^2}[\cos n\,\omega t + n\omega t \sin n\,\omega t]_0^T$$

$$= 0$$

Other Periodic Waves

$$B_n = \int_{\frac{-T}{2}}^{\frac{-T}{2}} f(t)\sin n\,\omega t dt = \int_0^T \frac{Vt}{T}\sin n\,\omega t dt = \frac{V}{T(n\omega)^2}\left[\sin n\,\omega t - n\omega t\cos n\,\omega t\right]_0^T$$

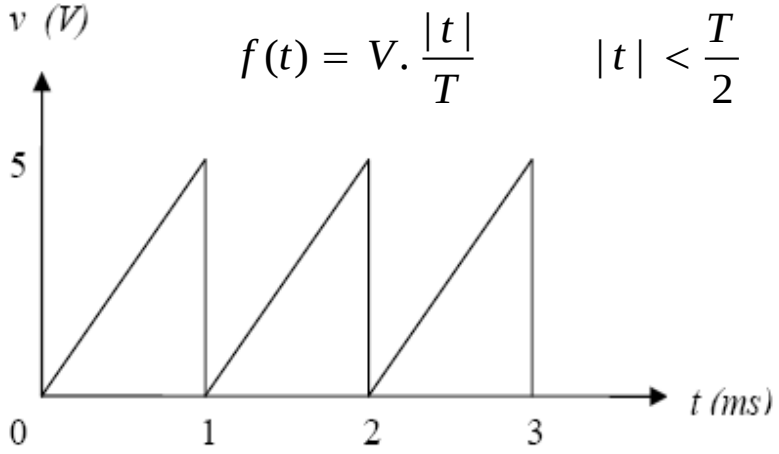
$$= \frac{-VT}{n2\pi}$$

$$f(t) = \frac{VT}{2T}.\frac{1}{T} + \frac{2}{T}\left(\frac{-VT}{2\pi}\right)\left[\sin n\omega t + \tfrac{1}{2}\sin 2\omega t + \tfrac{1}{3}\sin 3\omega t + \ldots\right]$$

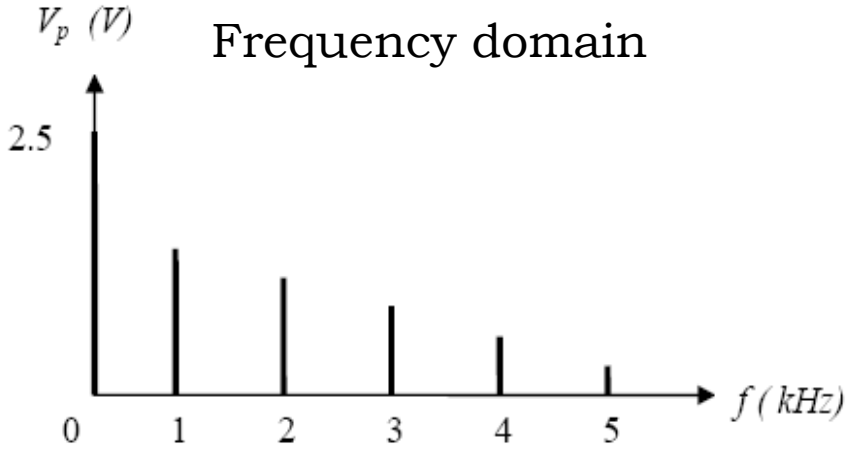
$$= \tfrac{V}{2} - \tfrac{V}{\pi}\left(\sin\omega t + \tfrac{1}{2}\sin 2\omega t + \tfrac{1}{3}\sin 3\omega t + \ldots\right)$$

Other Periodic Waves

- Positive going Sawtooth



Time domain



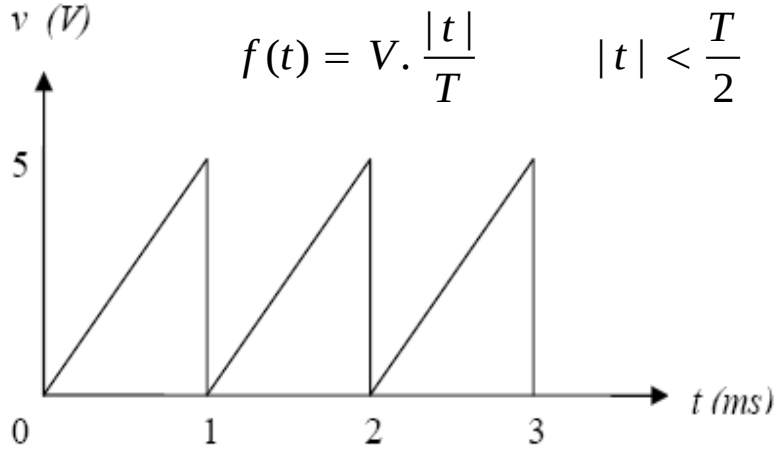
$f_0 = 1kHz; \quad Peak\ amplitude = 5V$

Fourier series

$$f(t) = \frac{V}{2} - \frac{V}{\pi}\left(\sin\omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \ldots\right)$$

Other Periodic Waves ../2

- Fourier series



Time domain

$$f(t) = \frac{V}{2} - \frac{V}{\pi}\left(\sin\omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \ldots\right)$$

$$= \frac{5}{2} - \frac{5}{\pi}\left(\sin 2\pi f\, t + \frac{1}{2}\sin 4\pi f\, t + \frac{1}{3}\sin 6\pi f\, t + \ldots\right)$$

$$= 2.5 - 1.59\left(\sin 2\pi\, x10^3\, t + \frac{1}{2}\sin 4\pi\, x\, 10^3\, t + \frac{1}{3}\sin 6\pi\, x10^3 t + \ldots\right)$$

Other Periodic Waves ../3

Example

- Determine bandwidth required to transmit the first seven components of a triangle wave with a fundamental frequency of 3 kHz
- *Solution*: 39 kHz

$$f(t) = \frac{8V}{\pi^2}\left(\cos\omega t + \frac{1}{3^2}\cos 3\omega t + \frac{1}{5^2}\cos 5\omega t + \frac{1}{7^2}\cos 7\omega t + \frac{1}{9^2}\cos 9\omega t + \frac{1}{11^2}\cos 11\omega t + \frac{1}{13^2}\cos 13\omega t + ..\right)$$

For first 7 components, highest freq. is given by 13th harmonic

Bandwidth $B = 13 f_0 = 13 x 3 = 39 kHz$

Other Periodic Waves ../4

Fourier-Series for selected waveforms

Periodic Waveform	Symmetry	Fourier Series
Square wave $f(t)=\begin{cases}+V & \lvert t\rvert<\frac{T}{4}\\ -V & \frac{T}{4}\le\lvert t\rvert<\frac{T}{2}\end{cases}$	Even	$f(t)=\frac{4V}{\pi}\left(\cos\omega t-\frac{1}{3}\cos 3\omega t+\frac{1}{5}\cos 5\omega t-\frac{1}{7}\cos 7\omega t+\cdots\right)$
Square wave $f(t)=\begin{cases}-V & -\frac{T}{2}<t<0\\ +V & 0\le t<\frac{T}{2}\end{cases}$	Odd	$f(t)=\frac{4V}{\pi}\left(\sin\omega t+\frac{1}{3}\sin 3\omega t+\frac{1}{5}\sin 5\omega t+\frac{1}{7}\sin 7\omega t+\cdots\right)$
Pulse train $f(t)=\begin{cases}+V & \lvert t\rvert<\frac{\tau}{2}\\ 0 & \frac{\tau}{2}\le\lvert t\rvert<\frac{T}{2}\end{cases}$	Even	$f(t)=\frac{V\tau}{T}+\frac{2V\tau}{T}\left(\frac{\sin\pi\tau/T}{\pi\tau/T}\cos\omega t+\frac{\sin 2\pi\tau/T}{2\pi\tau/T}\cos 2\omega t+\cdots\right)$
Triangle wave $f(t)=V\left(1-4\frac{\lvert t\rvert}{T}\right)\quad \lvert t\rvert<\frac{T}{2}$	Even	$f(t)=\frac{8V}{\pi^2}\left(\cos\omega t+\frac{1}{3^2}\cos 3\omega t+\frac{1}{5^2}\cos 5\omega t+\cdots\right)$

Other Periodic Waves ../5

Fourier-Series for selected waveforms

Half-wave Rectified sinusoid $f(t)=\begin{cases} V\sin\omega_o t & 0\le t<\frac{T}{2} \\ 0 & -\frac{T}{2}\le t<0 \end{cases}$	None	$f(t)=\frac{V}{\pi}+\frac{V}{2}\sin\omega t-\frac{2V}{\pi}\left(\frac{\cos 2\omega t}{1x3}+\frac{\cos 4\omega t}{3x5}+\cdots\right)$
Full-wave rectified sinusoid (Time zero at voltage zero) $f(t)=\lvert V\sin\omega_o t\rvert$	Even	$f(t)=\frac{2V}{\pi}-\frac{4V}{\pi}\left(\frac{\cos 2\omega t}{1x3}+\frac{\cos 4\omega t}{3x5}+\frac{\cos 6\omega t}{5x7}+\cdots\right)$
Full-wave rectified sinusoid (Time zero at voltage peak) $f(t)=\left\lvert V\sin\omega_o t+\frac{\pi}{2}\right\rvert$	Even	$f(t)=\frac{2V}{\pi}\left(1+\frac{2\cos 2\omega t}{1x3}-\frac{2\cos 4\omega t}{3x5}+\frac{2\cos 6\omega t}{5x7}-\cdots\right)$
Sawtooth (with no dc offset) $f(t)=2V\frac{t}{T}\qquad \lvert t\rvert<\frac{T}{2}$	Odd	$f(t)=\frac{2V}{\pi}\left(\sin\omega t-\frac{1}{2}\sin 2\omega t+\frac{1}{3}\sin 3\omega t-\frac{1}{4}\sin 4\omega t+\cdots\right)$
Sawtooth (positive-going) $f(t)=V\frac{\lvert t\rvert}{T}\qquad \lvert t\rvert<\frac{T}{2}$	None	$f(t)=\frac{V}{2}-\frac{V}{\pi}\left(\sin\omega t+\frac{1}{2}\sin 2\omega t+\frac{1}{3}\sin 3\omega t+\cdots\right)$

Fourier Transform

- Fourier-Series (FS)
  - Used for periodic signals
- Fourier Transform (FT)
  - Used for amplitude spectra of non-periodic signals
- Fourier transform may be developed from
  - exponential form of Fourier series by allowing period T to approach infinity.
  - For instance, a pulse with $T \rightarrow \infty$ means that $f(t)$ becomes pulse of width τ seconds

Fourier Transform ../2

i.e.

$$f(t) = \frac{1}{T}\sum_{-\infty}^{\infty} C_n \exp\left(j2\pi n f_0 t\right) \qquad \text{where } f_0 = \frac{1}{T} \;\&$$

$$C_n = \int_{-T/2}^{T/2} f(t)\exp\left(-j2\pi n f_0 t\right)dt$$

Let $\Delta f = (n+1)f_0 - nf_0 = \frac{1}{T}$ be spacing between harmonics of FS

- FS may be written as

$$f(t) = \sum_{-\infty}^{\infty} C_n \exp\left(j2\pi n f_0 t\right)\Delta f$$

Fourier Transform ../3

- As $T \rightarrow \infty$ then $\Delta f \rightarrow 0$ and

- Discrete harmonics in the series merge, and

- An amplitude spectrum that is a continuous function of frequency results, i.e.

$$\lim_{T \to \infty} C_n = \quad F(f)$$

Fourier Transform ../4

- The harmonic number *n* now has all possible values and
- summation of the series can be replaced by an integral,
- i.e. nf_0 is replaced by a continuous function *f* and

$$f(t) = \int_{-\infty}^{\infty} F(f)\exp\left(j2\pi nf_0 t\right)df$$

$$F(f) = \int_{-\infty}^{\infty} f(t)\exp\left(-j2\pi nf_0 t\right)dt$$

- The 2 integrals are known as FOURIER TRANSFORM PAIR.

Fourier Transform ../5

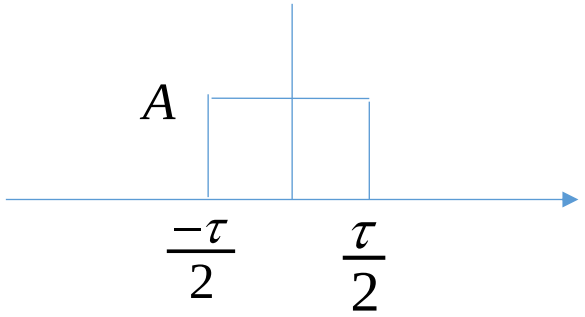
Example

- Assume $f(t)$ is a single pulse of amplitude A and duration τ seconds

$$F(f) = \int_{-\infty}^{\infty} f(t)\exp(-j2\pi n f_0 t)dt = \int_{-\tau/2}^{\tau/2} A\exp(-j2\pi f\, t)dt$$

$$= \frac{A}{\pi f}\sin(\pi\, f\tau)$$

$$= \frac{A\tau}{\pi\, f\tau}\sin(\pi f\tau) \;=\; A\tau\, Sinc(\pi f\tau)$$



- This type of function is called Sinc function of sampling function or SINE CARDINAL

Fourier Transform ../6

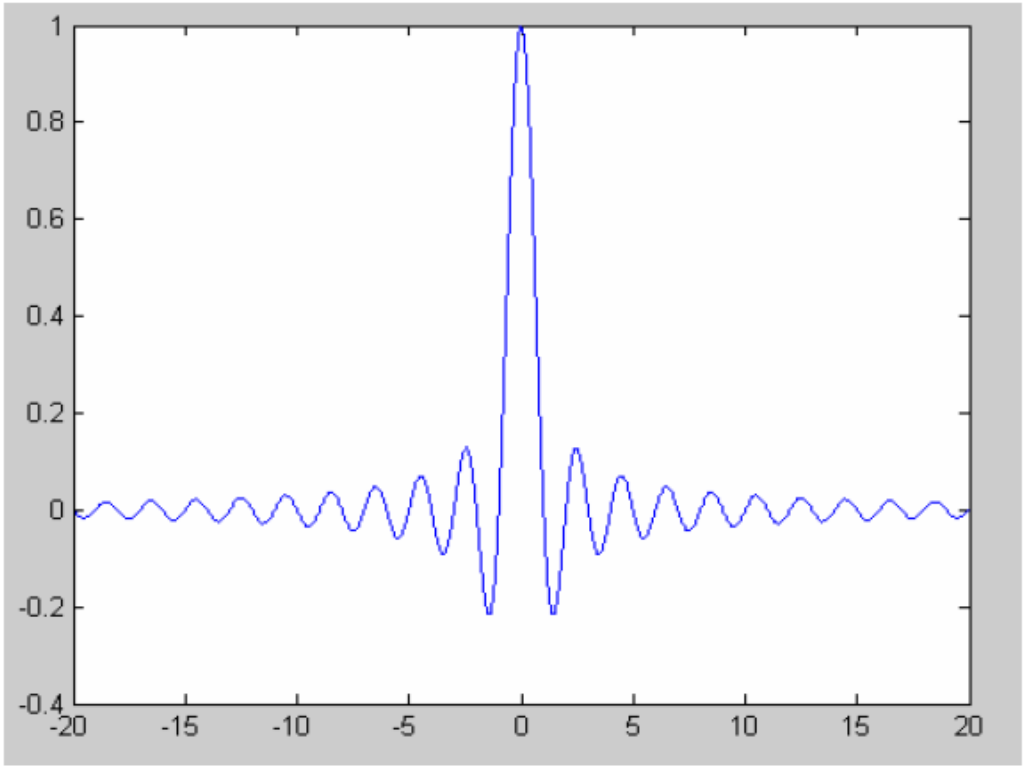
- Sinc function is defined as

$$Sinc\, x = \frac{\sin x}{x}$$

- Amplitude of Sinc function oscillates, decaying in either direction of x and approaches zero as $|x| \to \infty$

$$\left|\frac{\sin x}{x}\right| \to 0 \quad as \quad |x| \to \infty$$

$$\left|\frac{\sin x}{x}\right| \to \max value \quad as \quad |x| \to 0$$



Fourier Transform ../7

- Use L' Hopital Rule

$$If \quad f(x_0) = g(x_0) = 0$$

$$\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = \lambda, \qquad \text{where } \lambda \text{ is either real number or infinity}$$

then

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} = \lambda$$

It works for $x_0 = 0$ & $x_0 = \infty$

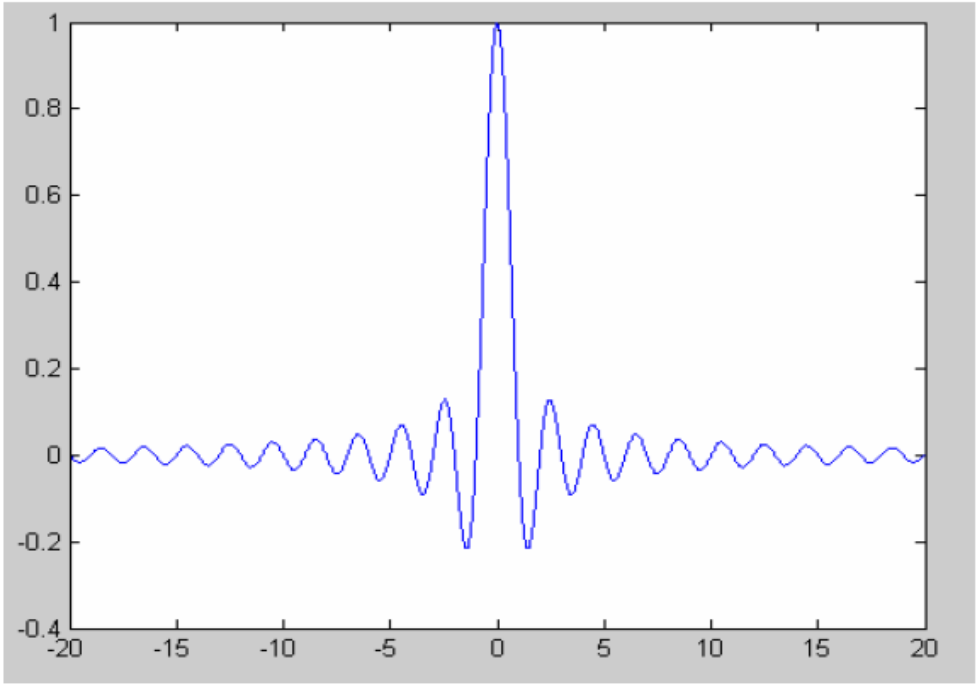
Fourier Transform ../7

- Using L' Hopital Rule

$$If \quad f(x) = \sin x, \quad and \quad g(x) = x$$

$$\lim_{x \to 0} \frac{f(x)}{g(x)} = \lim_{x \to 0} \frac{f'(x)}{g'(x)} = \lim_{x \to 0} \frac{\cos x}{1} = 1$$

$$\therefore \ \max \ value \ = 1$$



- envelope of this spectrum is a sinc function that has unity value when $\pi f \tau = 0$ &
- zero value when $\pi f \tau = m\,\pi$, i.e. when $f = m/\tau$

Fourier Transform ../7

- A rectangular pulse in time domain has a Fourier transform that is a sinc function in the frequency domain
- A sinc pulse in the time domain has a Fourier transform that is a rectangular function in the frequency domain